

Your grade: 90%

Your latest: 90% • Your highest: 90% • To pass you need at least 80%. We keep your highest score.

1.

Which application of xKF seeks to answer the question, "where am I?"

1 / 1 point

☐ Guidance.

☐ Target tracking.

☒ Navigation.

☐ State estimation.

☒ Correct

Yes, this is correct.
2.

In a navigation system that uses xKF, which of the following components fuses data from different sources to compute a position (and perhaps orientation) estimate?

1 / 1 point

☐ The gimbal system.

☒ The xKF itself.

☐ The INS.

☐ The GPS.

☒ Correct

Yes. That is correct.
3.

What is the principle of operation of triangulation?

1 / 1 point

☐ Multiple angle measurements between the sensor platform and precisely surveyed points are combined to estimate the orientation (angle) of the sensor platform.

☒ Multiple angle measurements between the sensor platform and precisely surveyed points are combined to estimate the position of the sensor platform.

☐ Multiple distance measurements between the sensor platform and precisely surveyed points are combined to estimate the orientation (angle) of the sensor platform.

☐ Multiple distance measurements between the sensor platform and precisely surveyed points are combined to estimate the position of the sensor platform.

☒ Correct

Yes, that is correct.
4.

What is the principle of operation of trilateration?

1 / 1 point

☐ Multiple angle measurements between the sensor platform and precisely surveyed points are combined to estimate the orientation (angle) of the sensor platform.

☒ Multiple distance measurements between the sensor platform and precisely surveyed points are combined to estimate the position of the sensor platform.

☐ Multiple distance measurements between the sensor platform and precisely surveyed points are combined to estimate the orientation (angle) of the sensor platform.

☐ Multiple angle measurements between the sensor platform and precisely surveyed points are combined to estimate the position of the sensor platform.

☒ Correct

Yes, that is correct.
5.

In a stochastic model of RSSI, what does the Rayleigh-fading component seek to describe?

1 / 1 point

☐ Loss in signal strength due to passing through an obstacle-free environment.

☐ Loss in signal strength due to obstructions in the path of propagation.

☒ Loss in signal strength due to multipath interference when there is no dominant line-of-sight path.

☐ Loss in signal strength due to multipath interference when there is a dominant line-of-sight path.

☒ Correct

Yes, this is correct.
6.

In a stochastic model of RSSI, what does the Rician-fading component seek to describe?

1 / 1 point

☐ Loss in signal strength due to passing through an obstacle-free environment.

☐ Loss in signal strength due to obstructions in the path of propagation.

☐ Loss in signal strength due to multipath interference when there is no dominant line-of-sight path.

☒ Loss in signal strength due to multipath interference when there is a dominant line-of-sight path.

☒ Correct

Yes, this is correct.
7.

If we cannot solve for $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ exactly in a particle filter, how might we approximate it?

1 / 1 point

☒ We could use the sigma-point method from Course 3 to find the mean and covariance of a Gaussian random variable that approximates $f(x_k^{(i)} \mid x_{k-1}^{(i)})$.

☐ We could assume that $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ is approximately equal to $f(z_k \mid x_k^{(i)})$, which we can implement.

☐ We can choose an arbitrary $q(x_k^{(i)} \mid x_{k-1}^{(i)})$ and set $f(x_k^{(i)} \mid x_{k-1}^{(i)}) = q(x_k^{(i)} \mid x_{k-1}^{(i)})$, equal to that. Then, the two distributions will cancel under the integral, making the precise nature of $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ irrelevant.

☐ We could assume that $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ is approximately equal to a Gaussian random variable having zero mean and covariance matrix equal to the identity matrix.

☒ Correct

Yes, this is correct. This is one of the methods that we implemented.
8.

Why might we choose the specific importance density $q(x_k^{(i)} \mid x_{k-1}^{(i)}, z_k) = f(x_k^{(i)} \mid x_{k-1}^{(i)})$ when implementing a particle filter?

1 point

☐ This is the only choice we can make. $q(x_k^{(i)} \mid x_{k-1}^{(i)}, z_k)$ is determined from the system model.

☐ Because it simplifies the weight-update computation.

☐ Because we do not know z_k at the point in the algorithm where we need to compute $q(x_k^{(i)} \mid x_{k-1}^{(i)}, z_k)$ and so we select an importance density that does not depend on z_k .

☒ Because it is always Gaussian and we know how to generate Gaussian random samples.

☒ Incorrect

No, this is not correct. In the example this week, it was Gaussian, but this is not generally true. See Lesson 4.4.4.
9.

Execute the Coursera Labs Jupyter notebook for Lesson 4.4.5. I recommend that you set the `showPlot` variable to zero to speed up the simulation (this will not change the result computed by the code; it will only cause the code not to plot the scatter plot).

1 / 1 point

Set the number of base stations to 1, and set the method for approximating $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ to `coordinates`. What is the root-mean-squared position error computed by the particle filter? Enter your answer with two digits to the right of the decimal point.

2.49

☒ Correct

Yes, that is correct.
10.

For the same Coursera Labs Jupyter notebook from Lesson 4.4.5, set the number of base stations to 2 and set the method for approximating $f(x_k^{(i)} \mid x_{k-1}^{(i)})$ to `coordinates`. Now, what is the root-mean-squared position error computed by the particle filter? Enter your answer with two digits to the right of the decimal point.

1 / 1 point

0.16

☒ Correct

Yes, that is correct.